CLASSROOM PLANETARY CLOCK & INTERVAL SYSTEM

Kinesthetic Time, Epicyclic 12:1 Gears & Modular Arithmetic

CARD 03 / 10 • TIME-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | **3·Clock** | 4·Fractions | 5·Primes | 6·Algebra | 7·Probability | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

EPICYCLIC 12:1 GEARS & MODULO ARITHMETIC

Time is continuous rotational motion & modular math.
StackCalc clock features an epicyclic gearset:
12:1 Exact Planetary Reduction Ratio
When minute hand turns 360°, hour hand advances 30°!

CALIBRATED 60-MINUTE / 12-HOUR DATUM

- Minute Unit Datum: 1 Minute = 6° dial sector
- Hour Mark: 1 Hour Mark = 30° arc (12 total sectors)
- Quadrant: 1/4 Hour Wedge = 90° right-angle sector
- Gear Ratio: 12.000:1 compound planetary gearset

CLOCK STAND LID & MODULAR WEDGE BAYS

Storage lid features easel stand & wedge bays.
Drop wedges into dial to measure elapsed time!

PLANETARY GEARS: 12:1 RATIO AT 3:30 (105°)



12:1 EPICYCLIC GEARS
360° min → 30° hour
At 3:30: Hour hand = 105°

15-Min Quadrant Wedge:
90° arc = 1/4 of an hour
Modulo 12 & Modulo 60

2. KIT PARTS INVENTORY (18 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Geared Clock Frame	1x	105.0 mm Dial (12h/60m)	Slate Gray
12:1 Epicyclic Gear	1x	Internal Sun-Planet Mesh	Gold / Blue
15-Min Wedges	4x	90° Quadrant (1/4 Hour)	Sky Blue
30-Min Wedges	2x	180° Semicircle (1/2 Hour)	Royal Purple
5-Min Wedges	8x	30° Hour Sector (1/12th)	Emerald Green
Display Stand Lid	1x	Clock Stand & Wedge Bay	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 16 Clock & Wedge Components + Storage Bin + Stand Lid = 18 Parts

FRACTION-ALIGNED TIME PALETTE

- Blue Wedges: 15-minute quadrants (90° right angles).
- Purple Wedges: 5-minute segments (30° hour steps).
- Emerald Face: Calibrated clock dial with 6° ticks.
- High-Visibility Hands: White minute & orange hour hands.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Seat the clock frame into its recessed bottom cradle. Drop the 14 colored wedges into their labeled radial lid bays. Any missing wedge leaves an empty bay. Easel lid latches firmly for silent, rattle-free carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

MODULO ARITHMETIC & CONTINUOUS RATIOS
Clock arithmetic is modulo 12 and modulo 60 governed by strict continuous gear kinematics.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 03 OF 10 • DOMAIN: KINEMATIC GEAR RATIOS & MODULAR ARITHMETIC • 12:1 EPICYCLIC REDUCTION • 15-MIN QUADRANT WEDGES

LESSON GUIDE: ELAPSED TIME & PLANETARY GEAR RATIOS

Step-by-Step Hands-On Tutorial: 3:30 + 0:45 = 4:15 & Modular Time

Grades 2–6 Elapsed Time & Modular Math • Hands-On Classroom & Home STEM Studio Series

CARD 03 / 10 • TIME-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/classroom-clock/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.3.MD.A.1 (Elapsed Time)
- CCSS.MATH.CONTENT.4.MD.A.2 (Time Intervals)
- TEKS 111.5.b.7.C (Elapsed Time Solutions)

Grades 2–6 Elapsed Time & Modular Math

💡 **MATHEMATICAL INQUIRY**

How far does the hour hand travel when the minute hand turns one full circle around the dial?

1 STEP 1: SET START TIME TO 3:30

Rotate the minute hand to 6 (bottom of dial, 30 minutes).
Observe the internal gears: the hour hand advances smoothly halfway between 3 and 4 (exactly 105° on the dial).

2 STEP 2: INSERT 15-MINUTE WEDGES

We need to add 45 minutes of elapsed time.
Insert three 15-minute wedges (3 × 15 min = 45 min).
Wedges cover 3 × 90° = 270° angular arc around dial.
Notice: The wedges bridge past the 12 o'clock mark!

3–4 STEPS 3 & 4: ADVANCE GEARS & READ RESULT

3. ADVANCE MINUTE HAND: Turn hand clockwise across the 3 wedges past 4:00 to 4:15!
4. READ FINAL TIME: Hour hand at 4:15; minute hand at 3 (15 min).
Calculator Verification: Press Clock HMS Icon ([🕒] / [g][+HMS] / [f][+HR]) and run 3.30 [f] [+HR] 0.45 [f] [+HR] [+] [g] [+HMS] = 4.15.

5 CORE PRINCIPLE: TIME IS CONTINUOUS KINEMATIC RATIO

Digital clocks make students think hour hands jump instantly when minutes reach 60. The planetary gears physically prove that time is a continuous ratio! Because 45 minutes is 3/4 of an hour, the hour hand moves through exactly 3/4 of its 30° sector (22.5°), landing precisely at 4:15!

★ EXTENSION INVESTIGATION: ELAPSED TIME: 8:45 + 1:30

Set clock to 8:45. Add two 45-minute wedges (90 min). Where does the planetary gear place the hour hand? (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

TIME-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 18 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California